

Body burden of persistent organic pollutants on hypertension: a meta-analysis

Su Hyun Park¹ · Jung-eun Lim^{2,3} · Hyesook Park¹ · Sun Ha Jee²

Received: 27 January 2016 / Accepted: 27 March 2016 / Published online: 7 April 2016
© Springer-Verlag Berlin Heidelberg 2016

Abstract Except the known risk factors for hypertension, several studies have suggested that exposure to persistent organic pollutants (POPs), endocrine disrupting chemicals, could be associated with an increased risk of hypertension. In this study, we performed a meta-analysis to summarize the existing epidemiological studies to investigate the association between POPs concentration and risk of hypertension. Based on comprehensive literature search results (PubMed, EMBASE, and KoreaMed), a meta-analysis of 11 articles was performed using a random-effects model. While we observed no significant association between the sum of non-dioxin-like PCBs and the risk of hypertension (OR = 1.00; 95 % CI 0.89, 1.12), the sum of dioxin-like PCBs was associated with a significantly increased risk of hypertension (OR = 1.45; 95 % CI 1.00, 2.12). High *p,p'*-DDE level was also significantly associated with the increased risk of hypertension. When subgroup analyses were carried out for studies

which analyzed POPs concentrations using high-resolution gas chromatography/high-resolution mass spectrometry, the overall pooled estimate ORs increased with decreased heterogeneity, providing it as a possible heterogeneity source (OR = 1.36; 95 % CI 1.21, 1.52; $I^2 = 42.8$ %; $p = 0.045$). This study suggested that the concentration of certain POPs, especially dioxin-related compounds, was associated with the risk of hypertension.

Keywords Blood pressure · Hypertension · Endocrine disruptors · Polychlorinated biphenyls

Introduction

The overall prevalence of hypertension among US adults aged 18 and over was 29.1 % in 2011–2012 (Nwankwo et al. 2013). The regulation of blood pressure is controlled by neurotransmitters and hormone, vascular resistance, and other factors (Goncharov et al. 2011). Known risk factors include obesity, sedentary lifestyle, high sodium intake, family history of hypertension, age, and tobacco use (Cook et al. 2007; Gus et al. 2004; Lin et al. 2009). Studies have suggested that exposure to endocrine disrupting chemicals could lead to chronic diseases including hormone-associated cancers and hypertension (Gao et al. 2015; Kreiss et al. 1981; Goncharov et al. 2010).

Persistent organic pollutants (POPs), environmental chemical compounds, are known endocrine disrupting chemicals, interfering with or mimicking the function of natural hormones. Polychlorinated biphenyls (PCBs) and *p,p'*-dichlorodiphenyldichloroethylene (*p,p'*-DDE) were used in agriculture and various industries until international environmental treaties were implemented in the 1990s to eliminate or

Responsible editor: Hongwen Sun

Su Hyun Park and Jung-eun Lim contributed equally to this work.

Electronic supplementary material The online version of this article (doi:10.1007/s11356-016-6568-6) contains supplementary material, which is available to authorized users.

✉ Sun Ha Jee
jsunha@yuhs.ac

¹ Department of Preventive Medicine, School of Medicine, Ewha Womans University, Seoul, Korea

² Institute for Health Promotion & Department of Epidemiology and Health Promotion, Graduate School of Public Health, Yonsei University, Seoul, Korea

³ Department of Public Health, Graduate School, Yonsei University, Seoul, Korea

restrict the production and use of them (Karlagnis et al. 2001). Substantial exposure has still been observed due to the persistency of these lipophilic compounds and affected human health (Longnecker et al. 2003; Iszatt et al. 2015).

An association between persistent organic pollutants (POPs) and hypertension has been controversial. Valera et al. (2013a, b) showed positive associations between polychlorinated biphenyls (PCBs) and hypertension, but various associations were observed in relation to organochlorine pesticides (OCPs). In PCBs, each congener showed different associations, and when divided PCBs by dioxin-like property, the results were not consistent between dioxin-like PCBs and nondioxin like PCBs (Valera et al. 2013b). Some studies showed positive, but no statistically significant, associations between POPs and hypertension (Ha et al. 2009; Lind et al. 2014). Therefore, we conducted a meta-analysis to provide possible clues on the association between POPs and hypertension risks and to confirm the effect of both congener specific and grouped together chemicals that share similar properties (dioxin like PCBs, nondioxin like PCBs, and OCPs) on hypertension with pooled data.

The objective of the present study was to summarize existing evidence on the association between POPs and hypertension risk in the general population by conducting a meta-analysis.

Methods

Search strategy

We conducted a literature search of PubMed (National Library of Medicine, Bethesda, MD) and Embase (Elsevier Publishers BV, Amsterdam) from the period 1949 through June 2015 to identify studies of POPs exposure and hypertension. Additionally, we searched a Korean medical database (KoreaMed) for access to Korean journals. The reference lists of obtained studies were also examined to identify other potential studies.

A total of 1796 related studies were selected using various combinations of the following keywords: “PCBs (polychlorinated biphenyls),” “OCPs (organochlorine pesticides),” “pesticides,” “hypertension,” “blood pressure,” “hypertensive,” “persistent organic pollutants,” “diabetes,” or “dioxin,” with no restriction on publication type, date, and the study language.

Two authors (SP and JL) independently screened the titles and abstracts of identified studies to determine the relevance to our research. The 57 studies were then examined in detail by full-text review to determine whether they met the inclusion criteria by two authors (SP and JL).

Inclusion and exclusion criteria

Studies were included if they fulfilled the following criteria: (1) The study should be an original article. Reviews, commentaries, conference abstracts, books, and meta-analyses were excluded. (2) The study should have direct measurements in human sample (blood serum, adipose tissue, etc.), rather than self-administered questionnaires of lifestyle and dietary intakes. Studies reporting on POPs levels from the environment (soil, air, water, etc.) were excluded. (3) Studies should be based on general population samples. For example, studies conducted in the high exposure areas or focused on elderly people were excluded. (4) Studies should be written in the English language. (5) Unavailable abstracts/full text were excluded.

After reviewing the full texts of 57 articles, 46 studies were excluded due to studies of irrelevant exposures/outcomes, unavailable full texts, non-English language publication, or insufficient study information. A total of 11 epidemiological studies (10 cross-sectional studies and 1 cohort study) were selected for the meta-analysis.

Data extraction and quality assessment

Data extraction was performed independently by the two researchers (SP and JL), after identifying all relevant articles. We collected the following information: author name, publication year, type of study, used sample, method, unit, number of subjects, adjustment variables, definition of hypertension, comparison categories, and effect size (ORs). When necessary, we emailed authors of original articles for additional data. We collected existing data of individual PCBs and OCPs of original articles to decide which and how many chemicals to be included in the meta-analysis. A total number of 47 POPs were collected: PCB 28, 52, 66, 74, 99, 101, 105, 118, 126, 128, 138, 146, 153, 156, 157, 158 and 138, 163, 169, 170, 172, 177, 178, 180, 183, 187, 189, 194, sum of PCBs, 2, 2',4,4'-tetrabromodiphenyl ether, polychlorinated dibenzo-*p*-dioxins (PCDDs), the sum of PCDDs, polychlorinated dibenzofurans (PCDFs), the sum of PCDFs, PCDDs/PCDFs, cis-nonachlordanes, trans-nonachlordanes, aldrin, mirex, heptachlor epoxide, octachlorodibenzodioxin, chlordane, hexachlorobenzene, hexachlorocyclohexane, oxychlordane, *p,p'*-DDT, *p,p'*-DDE, and sum of OCPs. A summary table of each POPs and references is presented (see Supplemental Material, Table S1).

Quality assessment was performed independently by two researchers (SP and JL) on 11 studies included in the meta-analysis (see Supplemental Material, Table S2), by using the Newcastle-Ottawa scale (Wells et al. 2009). The Newcastle-Ottawa scale for cross-sectional studies include three domains, with a maximum of nine points: selection, comparability, and outcome. We decided to include all 11 studies, since the

Newcastle-Ottawa scale scores of selected studies were equal to or greater than 6 (moderate-high quality). Any discrepancies in the rating of the study quality were resolved by a consensus.

Statistical analyses

A meta-analysis was conducted using Stata version 10.0 (StataCorp, College Station, TX, USA). The published odds ratio (OR) and 95 % confidence intervals (95 % CIs) were used for the measurement of association across studies. The random-effects model was used to consider both within- and between-study variations (DerSimonian and Laird 1986). Statistical heterogeneity among the selected studies was evaluated using the Q -test and I^2 statistic (Higgins et al. 2003), with $p < 0.05$ indicating statistical significance. The potential for publication bias was tested using Begg's (Begg and Mazumdar 1994) and Egger's test (Egger et al. 1997). Since three studies (Ha et al. 2009; Everett et al. 2008; Lee et al. 2007) used the National Health and Nutrition Examination Survey 1992–2002 data, we selected one with a larger sample size (Everett et al. 2008) for a meta-analysis. However, when we were not able to find the odds ratios for certain POPs, such as sum of dioxin-like PCBs, non-dioxin-like PCBs, and p,p' -DDE, we used data from another article (Lee et al. 2007). We conducted congener-specific meta-analyses, including five POPs: the dioxin-like PCB 118, the non-dioxin-like PCB 153, sum of dioxin-like PCBs, sum of non-dioxin-like PCBs, and one organochlorine pesticide— p,p' -DDE. Subgroup analyses were performed in order to address the effect of various characteristics on the pooled risk.

For the subgroup analyses in this study, we included studies based on the characteristics for each chemical,

such as whether (1) the POPs were reported on lipid-adjusted basis and (2) the POPs were measured by high-resolution gas chromatography/high-resolution mass spectrometry (HRGC/HRMS). Lastly, due to the various definitions of health outcomes, we selected studies for subgroup analysis when they met two of the following three criteria as a definition of hypertension: (a) history of physician or other health professional diagnosed, (b) use of antihypertensive medication treatment, and (c) measured systolic pressure ≥ 140 mmHg or measure or measured diastolic pressure ≥ 90 mmHg.

All analyses were performed using Stata version 10.0 software (StataCorp, College Station, TX). The null hypothesis of no difference was rejected if p values were less than 0.05.

Results

The flow of the literature search and selection process is shown in Fig. 1. A total of 1796 potentially relevant references were identified by a database search. After removal of duplicated publications, 1455 were screened for relevance on title and abstracts. Of the 57 references, 46 were excluded after full-text review. A total of 11 studies were selected in the meta-analysis.

The characteristics of studies included in the meta-analysis are summarized in Table 1. Four studies were conducted in the USA, two in Japan, one in Canada, one in Sweden, two in Spain, and one in Denmark. Except one cohort study, all studies were cross-sectional and used serum samples to analyze POPs. The POPs analyses of eight studies were performed with HRGC/HRMS, and nine studies reported the lipid-adjusted POPs. The variables adjusted among original studies

Fig. 1 Flow diagram of the study selection process

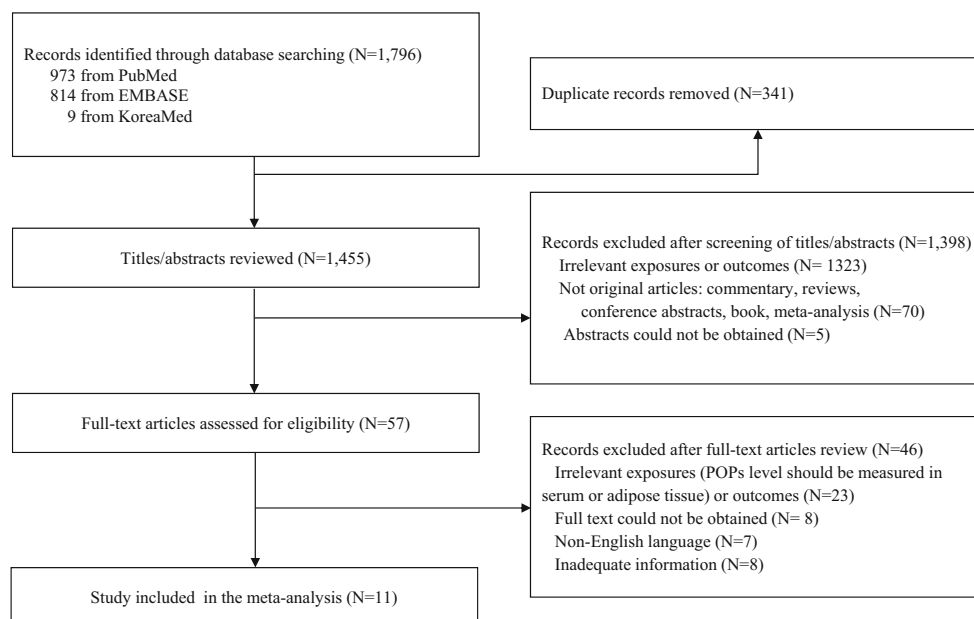


Table 1 Abstracted risk estimates and study information from the original studies relating serum POPs levels and hypertension

Reference	Country	Type of study	Used sample	Method	Unit	No. of subjects	Adjusting variables	Definition of chronic hypertension	Comparison categories	Effect size OR (95 % CI)
Lee et al. 2007	USA	Cross-sectional	Serum	HRGC-HRMS	ng/g lipid	721	Age, sex, race, poverty income ratio, cigarette smoking, serum cotinine, alcohol consumption, and exercise	SBP $\geq$ 130 mmHg or DBP $\geq$ 85 mmHg	Q4 vs. below the LOD	PCB 118: 1.8 (1.0–3.5) PCB 153: 0.8 (0.4–1.5) Σ DL-PCBs: 1.4 (0.8–2.7) Σ NDL-PCBs: 0.8 (0.4–1.5) p,p' -DDE: 1.3 (0.7–2.4)
Everett et al. 2008	USA	Cross-sectional	Serum	HRGC-HRMS	ng/g lipid	2074	Age, gender, race-ethnicity, smoking status, BMI, exercise, total cholesterol, and family history of heart attack	History of physician-diagnosed, use of medications for blood pressure, and SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg	T3 vs. T1	PCB118: 2.30 (1.29–4.08) PCB153: 1.45 (0.90–2.33)
Uemura et al. 2009	Japan	Cross-sectional	Serum	HRGC-HRMS	pg/g lipid	1374	Age, sex, smoking habit, drinking habit, regional block, residential area, and survey year	History of physician-diagnosed hypertension or SBP $\geq$ 130 mmHg or DBP $\geq$ 85 mmHg	Q4 vs. Q1	Σ DL-PCBs: 1.9 (1.1–3.1)
Ha et al. 2009	USA	Cross-sectional	Serum	HRGC-HRMS	ng/g lipid	524	Age, race, poverty income ratio, BMI, cigarette smoking, serum cotinine, alcohol consumption, and exercise.	SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg (treated hypertensives were excluded)	Q4 vs. below the LOD	In men, PCB118: 1.5 (0.5–4.3) Σ DL-PCBs: 2.3 (0.8–6.6) In women, PCB118: 0.3 (0.1–1.4) Σ DL-PCBs: 1.1 (0.3–3.5)
Yorita Christensen and White 2011	USA	Cross-sectional	Serum	HRGC-HRMS	ng/g	4,119	Age, sex, BMI, race/ethnicity, current smoking status, regular physical activity status, family history of cardiovascular diseases, total cholesterol, and serum lipid concentration	History of a doctor or other health professional- diagnosed or use of anti-hypertensive medication treatment or SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg	Q4 vs. Q1	PCB118: 1.63 (1.23–2.17) PCB153: 1.42 (1.03–1.95) Σ DL-PCBs: 1.26 (0.91–1.74)
Valera et al. 2013a, b (1)	Nunavik (Canada)	Cross-sectional	Serum	GC-ECD	μ g/L	315	Age, sex, fasting glucose, total lipids, waist circumference, alcohol consumption, smoking, and physical activity	Use of antihypertensive medication or SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg	per 1 unit increase	PCB118: 0.94 (0.79–1.12) PCB153: 1.02 (0.86–1.22) Σ DL-PCBs: 0.94 (0.78–1.13)

Table 1 (continued)

Reference	Country	Type of study	Used sample	Method	Unit	No. of subjects	Adjusting variables	Definition of chronic hypertension	Comparison categories	Effect size OR (95 % CI)
Valera et al. 2013a, b (2)	Greenland (Denmark)	Cross-sectional	Serum	GC-ECD	µg/kg lipid	1614	Age, sex, BMI, diabetes, physical activity, and smoking	Use of antihypertensive medication or SBP ≥140 mmHg or DBP ≥90 mmHg	Per 1 unit increase	ΣNDL-PCBs: 1.08 (0.90–1.29) <i>p,p'</i> -DDE: 1.10 (0.92–1.32) PCB118: 1.07 (0.93–1.23) PCB153: 0.96 (0.83–1.11) ΣDL-PCBs: 1.04 (0.89–1.21) ΣNDL-PCBs: 0.95 (0.81–1.11) ΣDL-PCBs: 2.4 (1.6–3.7)
Nakamoto et al. 2013	Japan	Cross-sectional	Serum	HRGC-MS	pg/g lipid	2264	Age, sex, smoking habit, drinking habit, regional block, survey year, and BMI	History of physician-diagnosed or SBP ≥140 mmHg or DBP ≥90 mmHg	Q4 vs. Q1	
Henriquez-Hernández et al. 2014	Spain	Cross-sectional	Serum	GC-ECD	µg/L	428	Age, gender, BMI, total lipids, aldrin, and smoking habits	Use of antihypertensive medication or SBP ≥140 mmHg or DBP ≥90 mmHg	Per 1 unit increase	<i>p,p'</i> -DDE: 1.04 (0.90–1.12)
Lind et al. 2014	Sweden	Cross-sectional	Serum	HRGC-HRMS	ng/g lipid	1012	Gender, BMI, smoking status, exercise habits, and education	Use of antihypertensive medication or SBP ≥140 mmHg or DBP ≥90 mmHg	Per 1 SD change	PCB118: 1.26 (0.95–1.67) PCB153: 1.19 (0.85–1.67) DDE: 1.25 (1.07–1.47)
Arrebola et al. 2015	Spain	Cohort	Adipose tissue	HRGC-MS	ng/g lipid	297	Age, BMI, smoking habit, and alcohol consumption	Receipt of antihypertensive therapy or SBP ≥140 mmHg or DBP ≥90 mmHg	Per 1 unit increase	PCB153: 1.13 (0.94–1.37) <i>p,p'</i> -DDE: 1.11 (0.93–1.33)

BMI body mass index, *DL* dioxin-like, *p,p'*-DDE 1,1-bis-(4-chlorophenyl)-2,2-dichloroethene, *p,p'*-DDT 1,1,1-trichloro-2,2-bis(4-chlorophenyl)ethane, *β-HCH* beta-hexachlorocyclohexane, *HR* high resolution, *GC-MS* gas chromatography–mass spectrometry, *ECD* electron capture detector, *OR* odds ratio, *NA* not available, *SBP* systolic blood pressure, *DBP* diastolic blood pressure, *LOD* lower limit of detection, *Q4* the highest quartile, *Q1* the lowest quartile, *T3* the highest tertile, *T1* the lowest tertile

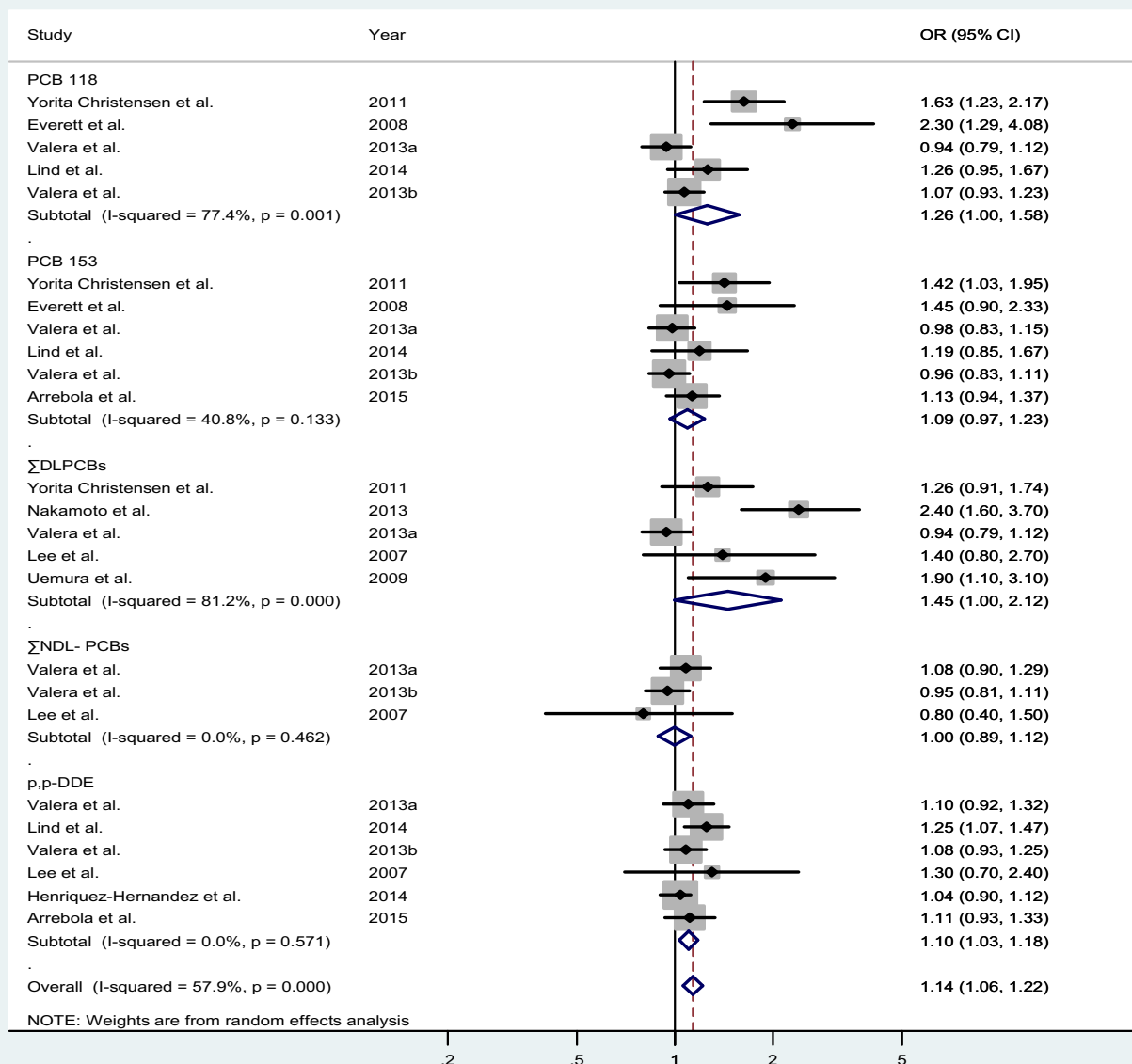


Fig. 2 Forest plot of pooled random-effects ORs (95 % CI) for the association between POPs concentrations and hypertension

varied across studies, but most studies included age, sex, socioeconomic status, BMI, smoking, alcohol consumption, and physical activity. The total number of subjects included in each study ranged from 297 to 4119. Six studies used the quantile categories, such as tertile (Everett et al. 2008) and quartile (Ha et al. 2009; Lee et al. 2007; Uemura et al. 2009; Yorita Christensen and White 2011; Nakamoto et al. 2013), whereas five studies (Valera et al. 2013a; Valera et al. 2013b; Lind et al. 2014; Henríquez-Hernández et al. 2014; Arrebola et al. 2015) presented a point estimate per one unit increase in POPs. The results presented that subjects in the highest category of POPs exposure were at the greatest risk of developing hypertension, compared with subjects in the lowest category, suggesting the linear dose–response relationship.

The data was pooled using the random-effect model, and the results presented that high concentrations of PCB118, the sum of dioxin-like PCBs, and *p,p'*-DDE were significantly associated with the risk of hypertension (Fig. 2). The pooled ORs of hypertension comparing highest versus lowest concentrations were 1.26 (95 % CI 1.00, 1.58) for PCB118, 1.45 (95 % CI 1.00, 2.12) for the sum of dioxin-like PCBs, and 1.10 (95 % CI 1.03, 1.18) for *p,p'*-DDE. We found positive associations between PCB 153 and the sum of non-dioxin-like PCBs, but none of these associations achieved statistical significance. Furthermore, there was a statistically significant positive association between total POPs of interest and hypertension (OR = 1.14, 95 % CI 1.06, 1.22).

The results of meta-analysis of all subgroups are summarized in Table 2. The heterogeneity statistics and pooled effect sizes of subgroups by lipid adjustment of concentrations and sharing the same outcome definitions for hypertension remained unchanged. When subgroup analyses were carried out for studies which analyzed POPs concentrations using HRGC/HRMS, the overall pooled estimate ORs increased with decreased heterogeneity, providing it as a possible heterogeneity source (OR = 1.36; 95 % CI 1.21, 1.52; $I^2 = 42.8\%$; $p = 0.045$). Specifically, the heterogeneity (I^2) for PCB 118 was reduced from 77.4 to 48.9 after subgroup analysis, and the magnitude of association with hypertension was found to be greater (OR = 1.56; 95 % CI 1.17, 2.07). For PCB 153, sum of dioxin-like PCBs and p,p' -DDE, we observed similar results, having increased the strength of association, as well as decreased heterogeneities.

Discussion

In this study, we conducted a meta-analysis to assess the association of POPs level identified in human biological samples with hypertension. When we selected and divided PCBs into two groups according to dioxin-like properties, this study

suggested that dioxin-like PCBs (PCB 118 and the sum of dioxin-like PCBs) were more strongly associated with hypertension than non-dioxin-like PCBs (PCB153 and the sum of non-dioxin-like PCBs). High concentrations of p,p' -DDE were also significantly associated with the increased risk of hypertension.

This meta-analysis was conducted on studies with POPs level identified in human biological samples to assess their associations with hypertension risk. Studies suggested that indirect measurement of POPs levels, including questionnaires or geographic information, might possibly lead to exposure misclassification (Kramer et al. 2012; Rushton 2003). Moreover, serum or adipose tissue has been used as proper human biological samples in epidemiological studies to investigate the association between POPs exposure and the risk of hypertension due to chemical properties such as long half-lives, lipophilicity, and bioaccumulation (Arrebola et al. 2015; Lind et al. 2014; Henríquez-Hernández et al. 2014).

Subgroup analyses were performed to explore the sources of heterogeneity of the original studies for subsets of studies (such as lipid-adjusted and lipid-unadjusted), for subsets of measurements (HRGC/HRMS or non-HRGC/HRMS), or for subsets of outcomes (specified definition of hypertension). The results of subsets of measurements presented increased

Table 2 Meta-analysis of the effect of the exposure to POPs on the risk of hypertension according to subgroup

Subgroup	No. of studies	Combined effect size	95 % CI	Weight	Q (p value)	I^2 (%)
PCB 118						
Lipid adjusted	3	1.31	(0.96, 1.80)	17.33	0.030	71.4
Definition of hypertension ^a	5	1.26	(1.00, 1.58)	20.80	0.001	77.4
HRGC-HRMS	3	1.56	(1.17, 2.07)	20.31	0.141	48.9
PCB 153						
Lipid adjusted	4	1.08	(0.94, 1.25)	24.91	0.227	30.8
Definition of hypertension ^a	6	1.09	(0.97, 1.23)	25.59	0.133	40.8
HRGC-HRMS	4	1.22	(1.06, 1.40)	31.11	0.566	0.0
ΣDL-PCBs						
Lipid adjusted	3	1.98	(1.47, 2.65)	7.54	0.353	3.9
Definition of hypertension ^a	3	1.38	(0.83, 2.27)	10.90	0.001	88.3
HRGC-HRMS	4	1.67	(1.21, 2.32)	19.07	0.099	52.2
ΣNDL-PCBs						
Lipid adjusted	2	0.94	(0.81, 1.10)	10.72	0.620	0.0
Definition of hypertension ^a	2	1.01	(0.89, 1.14)	11.73	0.293	9.4
p,p' -DDE						
Lipid adjusted	5	1.10	(1.03, 1.18)	39.50	0.427	0.0
Definition of hypertension ^a	5	1.10	(1.03, 1.17)	30.98	0.468	0.0
HRGC-HRMS	3	1.19	(1.06, 1.34)	29.52	0.598	0.0

p,p' -DDE 1,1-bis-(4-chlorophenyl)-2,2-dichloroethene, HR high resolution, GC-MS gas chromatography–mass spectrometry, DL-PCB dioxin-like PCBs, NDL-PCBs, non-dioxin-like PCBs.

^a Studies were included if hypertension was defined as meeting two of the following three criteria: (1) history of physician or other health professional diagnosed, (2) use of antihypertensive medication treatment, and (3) measured systolic pressure ≥ 140 mmHg or measure or measured diastolic pressure ≥ 90 mmHg

overall pooled OR with decreased heterogeneity (see Supplemental Material, Figure S1). We divided the original studies in subgroups that implemented different analytical methods, since different methods might yield different results. The HRGC/HRMS method is the most widely used and recommended for POPs analyses as the most promising analytical technique due to its high sensitivity and selectivity, as well as low limits of detection (Judd et al. 2003). The US Environmental Protection Agency (EPA) (USEPA 1994; USEPA 2010) and European Union (EU) regulations require HRGC/HRMS measurement for dioxin analysis (EU 2011; EU 2012). Since quantifying exposure to environmental chemicals using biologic samples requires expensive, complex, and time-consuming methods, summarizing effect sizes from original studies that used the same analytical method might reduce heterogeneity across studies.

This meta-analysis was conducted targeting the most well-known POPs: dioxin-like PCB 118, non-dioxin-like PCB 153, sum of dioxin-like PCBs, sum of non-dioxin-like PCBs, and *p,p'*-DDE. In addition to these compounds, POPs contain other compounds, such as polybrominated diphenyl ethers (PBDEs) and perfluorinated compounds (PFC). However, we could not perform meta-analysis of these compounds because of the insufficient number of previously published epidemiologic studies that observed the association between these compounds and the risk of hypertension. A study reported that there was no statistically significant association between PBDEs and gestational hypertension (Smarr et al. 2016). Yet, a positive association of perfluorooctanoic acid (PFOA) or perfluorooctane sulfonate (PFOS) levels and serum uric acid levels, a marker of the increased risk of developing hypertension, was observed (Steenland et al. 2010; Shankar et al. 2006). A cross-sectional study using data from the National Health and Nutrition Examination Survey (NHANES) 1999–2002 suggested that serum concentrations of polychlorinated dibenzo-*p*-dioxins (PCDDs) and polychlorinated dibenzofurans (PCDFs) were associated with increased prevalence of newly diagnosed hypertension among women, but not among men (Ha et al. 2009). The authors explained the stronger positive association of PCDDs or PCDFs with hypertension among women due to the antiestrogenic effects of aryl hydrocarbon receptor (AhR) agonists.

To our knowledge, this is the first study to comprehensively assess the association between POPs levels identified in human biological samples and hypertension. In this meta-analysis, we assessed the quality of included studies to explore other potential sources of bias. All studies included in the meta-analysis were conducted using the same study design and commonly used serum POPs levels, except one study (Arrebola et al. 2015). This could increase the homogeneity of meta-analysis. After various subgroup analyses, we found that the method used to measure POPs level could be an

important factor that increases the homogeneity between studies. Lastly, when we assessed the associations between PCBs and hypertension, we grouped PCB congeners as dioxin-like or non-dioxin-like. Among PCBs, dioxin-like PCBs were highly associated with hypertension risk.

Several limitations need to be discussed. First, overall interpretation of results requires a reserved caution, and further prospective studies with long-term follow-up are needed to validate and confirm these results, since we summarized the data from cross-sectional studies by conducting a meta-analysis. However, the persistence property of POPs could reduce the possibility of reverse causation, because the measured concentrations could reflect comparably long-term exposure amounts of POPs. Second, we tested linear relationships between total PCBs and risk of hypertension in this meta-analysis. However, some studies suggested inverted *U*-shape or monotonically increasing association within the low-dose exposure range of POPs with chronic disease (Daston et al. 2003; Welshons et al. 2003; Lee et al. 2014). More research should be granted to clarify this complex dose–response relationship. Third, although we used a random-effects model and performed subgroup analyses to take into account the possible presence of heterogeneity, heterogeneity could still exist due to methodological differences between studies. Fourth, because POPs are intercorrelated, we cannot exclude the possibility that the observed significant associations in PCB 118, the sum of dioxin-like PCBs, and *p,p*-DDE may actually be due to the effects of individual chemicals, or other POPs. Fifth, there is a possibility of publication and selection bias due to the inclusion of only published studies and English-language studies, even though we conducted a comprehensive search strategy to minimize it. Although funnel plot was applied, it was difficult to interpret because less than 10 studies were included for a meta-analysis of each POPs. Generally, at least 10 studies in the meta-analysis would be required to identify for publication bias using the funnel plot (Higgins and Green 2011). *p,p*-DDE, reported by the largest number of studies, was selected to assess publication bias, and the results from Egger's and Begg's tests were not significant ($p=0.30, 0.26$, respectively).

The mechanisms responsible for the relationship between POPs and hypertension have not been clearly elucidated. Experimental evidence suggests that PCBs can modulate endothelium-derived vasoactive factors in human primary endothelial cells and induce changes in human endothelial cells, which is characteristic of endothelial dysfunction in human hypertension (Andersson et al. 2011). To be specific, PCBs can especially increase the expression of vasoconstriction factors, such as cyclooxygenase (COX-2), prostaglandins, and reactive oxygen species (ROS), which modify the increase in blood pressure (Andersson et al. 2011). In addition, PCB levels decrease nitric oxide (NO) production, a vasodilator of the microcirculation (Andersson et al. 2011). Xu et al. (2006)

reported that *p,p'*-DDE is a kind of anti-androgen compound which acts on the androgen receptor, and recently, *p,p'*-DDE levels have been found to be inversely associated with testosterone levels in men (Xu et al. 2006; Blanco-Muñoz et al. 2012). Since low testosterone levels in men have been associated with hypertension (Akishita et al. 2010; Svartberg et al. 2004), anti-androgen activity is considered as a plausible mechanism whereby *p,p'*-DDE might affect blood pressure (Lind et al. 2014). Moreover, animal studies showed that one of the dioxin-like chemicals, 2,3,7,8-tetrachlorodibenzo-*p*-dioxin, may increase blood pressure via increased renal oxidative stress and vascular reactivity (İlhan et al. 2015). This could explain the results of our study concluding that dioxin-like chemicals could be more associated with increased blood pressure than non-dioxin-like chemicals.

In conclusion, we investigated the association between POPs level and hypertension, and different results were presented depending on the individual compounds and mixtures. This study suggested that continued exposure to POPs, especially dioxin-related compounds, might be associated with an increased risk of hypertension. Further epidemiological studies to confirm the hypertensive effect of emerging POPs, such as PBDEs or PFCs, are needed.

Acknowledgments This research was supported by a grant (13162MFDS891) from the Ministry of Food and Drug Safety in 2013 and a grant (15162MFDS631) from the Ministry of Food and Drug Safety in 2015.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

References

- Akishita M, Fukai S, Hashimoto M, Kameyama Y, Nomura K, Nakamura T et al (2010) Association of low testosterone with metabolic syndrome and its components in middle-aged Japanese men. *Hypertens Res* 33(6):587–591
- Andersson H, Garscha U, Brittebo E (2011) Effects of PCB126 and 17beta-oestradiol on endothelium-derived vasoactive factors in human endothelial cells. *Toxicology* 285(1–2):46–56
- Arrebola JP, Fernández MF, Martín-Olmedo P, Bonde JP, Martín-Rodríguez JL, Expósito J, Rubio-Domínguez A, Olea N (2015) Historical exposure to persistent organic pollutants and risk of incident hypertension. *Environ Res* 138:217–223
- Begg CB, Mazumdar M (1994) Operating characteristics of a rank correlation test for publication bias. *Biometrics* 50:1088–1101
- Blanco-Muñoz J, Lacasaña M, Aguilar-Garduño C, Rodríguez-Barranco M, Bassol S, Cebrián ME et al (2012) Effect of exposure to *p,p'*-DDE on male hormone profile in Mexican flower growers. *Occup Environ Med* 69(1):5–11
- Cook NR, Cutler JA, Obarzanek E, Buring JE, Rexrode KM, Kumanyika SK et al (2007) Long term effects of dietary sodium reduction on cardiovascular disease outcomes: observational follow-up of the trials of hypertension prevention (TOHP). *BMJ* 334(7599):885–888
- Daston GP, Cook JC, Kavlock RJ (2003) Uncertainties for endocrine disruptors: our view on progress. *Toxicol Sci* 74(2):245–252
- DerSimonian R, Laird N (1986) Meta-analysis in clinical trials. *Control Clin Trials* 7(3):177–188
- Egger M, Davey Smith G, Schneider M, Minder C (1997) Bias in metaanalysis detected by a simple, graphical test. *BMJ* 315:629–634
- Everett CJ, Mainous AG 3rd, Frithsen IL, Player MS, Matheson EM (2008) Association of polychlorinated biphenyls with hypertension in the 1999–2002 National Health and Nutrition Examination Survey. *Environ Res* 108(1):94–97
- Gao H, Yang BJ, Li N, Feng LM, Shi XY, Zhao WH, Liu SJ (2015) Bisphenol A and hormone-associated cancers: current progress and perspectives. *Medicine (Baltimore)* 94(1), E211
- Goncharov A, Bloom M, Pavuk M, Birman I, Carpenter DO (2010) Blood pressure and hypertension in relation to levels of serum polychlorinated biphenyls in residents of Anniston, Alabama. *J Hypertens* 28(10):2053–2060
- Goncharov A, Pavuk M, Foushee HR, Carpenter DO (2011) Blood pressure in relation to concentrations of PCB congeners and chlorinated pesticides. *Environ Health Perspect* 119(3):319–325
- Gus M, Fuchs SC, Moreira LB, Moraes RS, Wiehe M, Silva AF et al (2004) Association between different measurements of obesity and the incidence of hypertension. *Am J Hypertens* 17:50–53
- Ha MH, Lee DH, Son HK, Jacobs DR Jr (2009) Association between serum concentrations of persistent organic pollutants and prevalence of newly diagnosed hypertension: results from the National Health and Nutrition Examination Survey 1999–2002. *J Hum Hypertens* 23(4):274–286
- Henríquez-Hernández LA, Luzardo OP, Zumbado M, Camacho M, Serra-Majem L, Álvarez-León EE, Boada LD (2014) Blood pressure in relation to contamination by polychlorobiphenyls and organochlorine pesticides: results from a population-based study in the Canary Islands (Spain). *Environ Res* 135:48–54
- Higgins JPT, Green S (2011) Cochrane handbook for systematic reviews of interventions, version 5.1.0. <http://www.cochranehandbook.org>. Accessed 20 Mar 2015
- Higgins JP, Thompson SG, Deeks JJ, Altman DG (2003) Measuring inconsistency in meta-analyses. *BMJ* 327(7414):557–560
- İlhan S, Ateşşahin D, Ateşşahin A, Mutlu E, Onat E, Sahna E (2015) 2,3,7,8-Tetrachlorodibenzo-*p*-dioxin-induced hypertension: the beneficial effects of melatonin. *Toxicol Ind Health* 31(4):298–303
- Iszatt N, Stigum H, Verner MA, White RA, Govarts E, Palkovicova Murinova L et al (2015) Prenatal and postnatal exposure to persistent organic pollutants and infant growth: a pooled analysis of seven European birth cohorts. *Environ Health Perspect* 123(7):730–736
- Judd NL, Griffith WC, Kalman DA, Faustman EM (2003) Assessment of PCB congener analytical methods Do they meet risk assessment needs? *Arch Environ Contam Toxicol* 44(1):132–139
- Karlaganis G, Marioni R, Sieber I, Weber A (2001) The elaboration of the 'Stockholm convention' on persistent organic pollutants (POPs): a negotiation process fraught with obstacles and opportunities. *Environ Sci Pollut Res Int* 8(3):216–221
- Kramer S, Hikel SM, Adams K, Hinds D, Moon K (2012) Current Status of the Epidemiologic Evidence Linking Polychlorinated Biphenyls and Non-Hodgkin Lymphoma, and the Role of Immune Dysregulation. *Environ Health Perspect* 120:1067–1075
- Kreiss K, Zack MM, Kimbrough RD, Needham LL, Smrek AL, Jones BT (1981) Association of blood pressure and polychlorinated biphenyl levels. *JAMA* 245(24):2505–2509
- Lee DH, Lee IK, Porta M, Steffes M, Jacobs DR Jr (2007) Relationship between serum concentrations of persistent organic pollutants and the prevalence of metabolic syndrome among non-diabetic adults: results from the National Health and Nutrition Examination Survey 1999–2002. *Diabetologia* 50(9):1841–1851

- Lee DH, Porta M, Jacobs DR Jr, Vandenberg LN (2014) Chlorinated persistent organic pollutants, obesity, and type 2 diabetes. *Endocr Rev* 35(4):557–601
- Lin CY, Chen PC, Kuo HK, Lin LY, Lin JW, Hwang JJ (2009) Effects of obesity, physical activity, and cardiorespiratory fitness on blood pressure, inflammation, and insulin resistance in the National Health and Nutrition Survey 1999–2002. *Nutr Metab Cardiovasc Dis* 20(10):713–719
- Lind PM, Penell J, Salihovic S, van Bavel B, Lind L (2014) Circulating levels of p, p'-DDE are related to prevalent hypertension in the elderly. *Environ Res* 129:27–31
- Longnecker MP, Korrick SA, Moysich KB (2003) *Dioxins and Health*, 2nd edn. John Wiley & Sons, Inc., New Jersey, pp 679–728
- Nakamoto M, Arisawa K, Uemura H, Katsuura S, Takami H, Sawachika F et al (2013) Association between blood levels of PCDDs/PCDFs/dioxin-like PCBs and history of allergic and other diseases in the Japanese population. *Int Arch Occup Environ Health* 86(8):849–859
- Nwankwo T, Yoon SS, Burt V, Gu Q (2013) Hypertension among adults in the United States: National Health and Nutrition Examination Survey, 2011–2012. *NCHS Data Brief* 133:1–138
- Rushton L (2003) Health hazards and waste management. *Br Med Bull* 68:183–197
- Shankar A, Klein R, Klein BE et al (2006) The association between serum uric acid level and long-term incidence of hypertension: population-based cohort study. *J Hum Hypertens* 20(12):937–945
- Smarr MM, Grantz KL, Zhang C, Sundaram R, Maisog JM, Barr DB, Louis GM (2016) Persistent organic pollutants and pregnancy complications. *Sci Total Environ* 551–552:285–291
- Steenland K, Tinker S, Shankar A et al (2010) Association of perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS) with uric acid among adults with elevated community exposure to PFOA. *Environ Health Perspect* 118(2):229–333
- Svartberg J, Jenssen T, Sundsfjord J, Jorde R (2004) The associations of endogenous testosterone and sex hormone-binding globulin with glycosylated hemoglobin levels, in community dwelling men. The Tromsø Study. *Diabetes Metab* 30(1):29–34
- U.S. EPA (Environmental Protection Agency) (1994) Method 1613, Tetra-through Octa-Chlorinated Dioxins and Furans by Isotope Dilution HRGC/HRMS. http://www.well-labs.com/docs/epa%20method%201613_oct1994.pdf. Accessed 12 Jan 2016
- U.S. EPA (Environmental Protection Agency) (2010) Method 1668C, Chlorinated Biphenyl Congeners in Water, Soil, Sediment, Biosolids, and Tissue by HRGC/HRMS. http://water.epa.gov/scitech/methods/cwa/upload/M1668C_11June10-PCB_Congeners.pdf. Accessed 12 Jan 2016
- Uemura H, Arisawa K, Hiyoshi M, Kitayama A, Takami H, Sawachika F et al (2009) Prevalence of metabolic syndrome associated with body burden levels of dioxin and related compounds among Japan's general population. *Environ Health Perspect* 117(4):568–573
- EU (European Union) (2011) Commission Regulation (EU) NO 1259/2011 of 2 December 2011 amending Regulation (EC) No 1881/2006 as regards maximum levels for dioxins, dioxin-like PCBs and non dioxin-like PCBs in foodstuffs. *Off J Eur Union*. L320: 18–22. <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2011:320:0018:0023:EN:PDF>. Accessed 12 Jan 2016
- EU (European Union) (2012) Commission Regulation (EU) NO 252/2012 of 21 March 2012 laying down methods of sampling and analysis for the official control of levels of dioxins, dioxin-like PCBs and non-dioxin-like PCBs in certain foodstuffs and repealing Regulation (EC) No 1882/2006. *Off J Eur Union*. L84:1–22. https://www.fsai.ie/uploadedfiles/reg252_2012.pdf. Accessed 12 Jan 2016
- Valera B, Ayotte P, Poirier P, Dewailly E (2013a) Associations between plasma persistent organic pollutant levels and blood pressure in Inuit adults from Nunavik. *Environ Int* 59:282–289
- Valera B, Jørgensen ME, Jeppesen C, Bjerregaard P (2013b) Exposure to persistent organic pollutants and risk of hypertension among Inuit from Greenland. *Environ Res* 122:65–73
- Wells G, Shea B, O'Connell D, Peterson J, Welch V, Losos M, et al. (2009) The Newcastle–Ottawa Scale (NOS) for assessing the quality if nonrandomized studies in meta-analyses; http://www.ohri.ca/programs/clinical_epidemiology/oxford.htm [Accessed 27 Jan 2016]
- Welshons WV, Thayer KA, Judy BM, Taylor JA, Curran EM, vom Saal FS (2003) Large effects from small exposures. I. Mechanisms for endocrine-disrupting chemicals with estrogenic activity. *Environ Health Perspect* 111(8):994–1006
- Xu LC, Sun H, Chen JF, Bian Q, Song L, Wang XR (2006) Androgen receptor activities of p, p'-DDE, fenvalerate and phoxim detected by androgen receptor reporter gene assay. *Toxicol Lett* 160(2):151–157
- Yorita Christensen KL, White P (2011) A methodological approach to assessing the health impact of environmental chemical mixtures: PCBs and hypertension in the National Health and Nutrition Examination Survey. *Int J Environ Res Public Health* 8(11):4220–4237